

Computer Forensics

CTMSDFIS SI P1

Niharika Pagare

CTMSDFIS SI P1: Computer Forensics

Teaching Scheme					Evaluation Scheme							
Th	Tu	Pr	C	TCH	Theory				Practical			
					Internal Exams		University Exams		University Exams (LPW)		Total	
					TA-1/TA-2	MSE	Marks	Hrs	Marks	Hrs		
G3	06	00	03	03	25	00:45	50	01:30	100	03:00	-	200

* Note: TA-2 will be in the form of assignments or workshops.

1. To learn the process of recovering, analyzing, and preserving computer and related materials in such a manner that it helps the investigation agency to present them as evidence in a court of law.
2. To postulate the motive behind the crime and identity of the main culprit.
3. Designing procedures at a suspected crime scene which can help in ensuring that the digital evidence obtained is not corrupted.
4. Data acquisition and duplication: Recovering deleted files and deleted partitions from digital media to extract the evidence and validate

UNIT – I

Fundamentals of Computer Systems

Number Systems: Binary, Octal, Decimal and Hexadecimal, Conversion and Operations on Binary Number Systems, Representing Information in Hexadecimal and Binary, Little Endian and Big-Endian Formats. **Computer Architecture:** Input Devices, Central Processing Unit, Storage Devices and Output Devices. **Secondary Storage Devices:** Understanding Disk, Volume and Partition, HDD and SSD Structures, Introduction to MBR, GPT and VBR. **Basics of Operating Systems:** Role of Operating System, Boot Process, Introduction to Process and Memory Management

UNIT – II

Introduction to Computer Forensics Investigations and Electronic Evidence

Digital Forensics: Definition, Process, Locard's Principle of Exchange, Branches of Digital Forensics, Handling Digital Crime Scene, Important Documents and Electronic Evidences. **Introduction to Evidence Acquisition:** Identification, Acquisition, Labeling and Packaging, Transportation, Chain-of-Custody, Importance of Documentation and Preservation, Acquisition Process: Write-Blockers, Imaging Techniques, Evidence Integrity, Standard Operating Procedures for Acquisitions and Preservation of Evidences. **Introduction to Data Recovery and Carving:** Importance of Data Recovery in Forensic Investigation, Carving Methods, Difference between Data Recovery and Carving

Unit - III

Windows OS Architecture - File System Analysis: Understanding and Analyzing FAT and NTFS File Systems, Recreating FAT and NTFS Partitions, Analyzing Unallocated Partitions. **Registry Analysis:** Understanding Windows Registry, Analyzing Windows Registry, Finding Important Artefacts Related to User Activities, User/Application Configurations and Preferences; Attached Devices, Shared Locations, Recently Accessed Documents, Programs and Locations; Installed Applications and Others from Windows Registry. **Event and Log Analysis:** Introduction to Windows Events, Understanding Windows Events (Evt and Evtx Files). Analyzing Logs of Third-party Applications.

UNIT – IV

Linux OS Forensics

Linux OS Architecture - File System Analysis: Understanding and Analyzing EXT File System, Recreating EXT Partitions, Analyzing Unallocated Partitions. **Log Analysis:** Understanding Various Logs, Analyzing Important Logs to Find Important Artefacts Related to User Activities, User/Application Configurations and Preferences; Attached Devices, Shared Locations, Recently Accessed Documents, Programs and Locations; Installed Applications.

UNIT – V

MAC OS Forensics

MAC OS Architecture - File System Analysis: Understanding and Analyzing HFS and HFS+ File System, Recreating HFS Partitions, Analyzing Unallocated Partitions, Understanding APFS. **Log Analysis:** Understanding Various Logs, Analyzing Important Logs to Find Important Artefacts Related to User Activities, User/Application Configurations and Preferences; Attached Devices, Shared Locations, Recently Accessed Documents, Programs and Locations; Installed Applications.

Reference Books: -

1. **Practical Guide to Computer Forensics Investigations, A (Pearson IT Cybersecurity Curriculum (ITCC))** 1st Edition by Darren R. Hayes
2. **Learn Computer Forensics: A beginner's guide to searching, analyzing, and securing digital evidence** 1st Edition by William Oettinger
3. **Investigating Windows Systems** 1st Edition by Harlan Carvey.
4. **Practical Linux Forensics: A Guide for Digital Investigators** by Bruce Nikkel
5. **Linux Forensics** Kindle Edition by Philip Polstra
6. **Computer Systems_ Digital Design, Fundamentals of Computer Architecture and Assembly Language**
7. **Digital Forensics with Open-Source Tools** 1st Edition by Cory Altheide, Harlan Carvey
8. **Operating Systems: Internals and Design Principles, 9/e** by William Stallings
9. **Linux Bible 10th Edition** by Christopher Negus
10. **Linux Pocket Guide: Essential Commands** 3rd Edition by Daniel J. Barrett

Unit 1: Fundamentals of Computer System [20 marks]

Computer Architecture: Input devices, CPU, Storage Devices and Output devices

Secondary storage devices: understanding disk, volume and partition, HDD & SSD structures, introduction to MBR, GPT and VBR

Basics of Operating System: Role of OS, Boot process, Introduction to Process and Memory Management

Number System: Binary, Decimal, Octal and Hexadecimal, Conversions and Operations on Binary Number System, Representing Information in Hexadecimal and Binary, Little Endian and Big-Endian Formats.

Big Picture

Computer architecture relies on the “**rules and design**” that define how a **CPU works**.

- It includes:
 - **Instruction Set Architecture (ISA):** the “language” of the CPU.
 - **Microarchitecture:** how the CPU internally executes the ISA.
 - **System design:** memory, I/O, buses, etc.

- **x86 (Intel/AMD)**
 - Dominates PCs, laptops, and most servers.
 - Strength: **high performance, backward compatibility**.
 - Weakness: **power-hungry, complex instruction set**.
- **ARM (ARM Holdings)**
 - Dominates mobiles, tablets, IoT, and increasingly laptops/servers.
 - Strength: **power efficiency, small/simple instruction set (RISC)**.
 - Weakness: traditionally lower performance (but catching up).

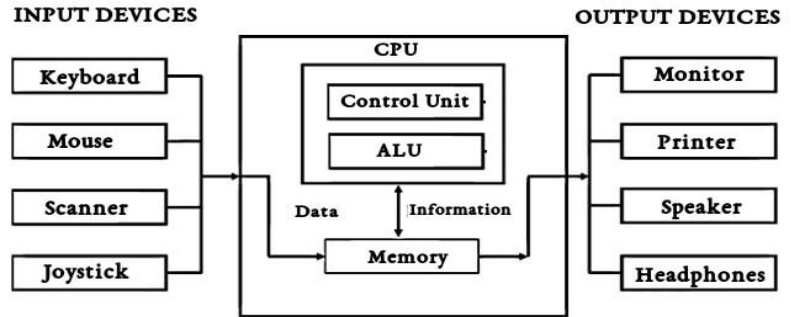
So,

- “Your Windows laptop likely uses **x86** (Intel/AMD).”
- “Your smartphone uses **ARM** (Qualcomm, Apple, Samsung chips).”
- “The cloud may use **both**, depending on cost and efficiency.”

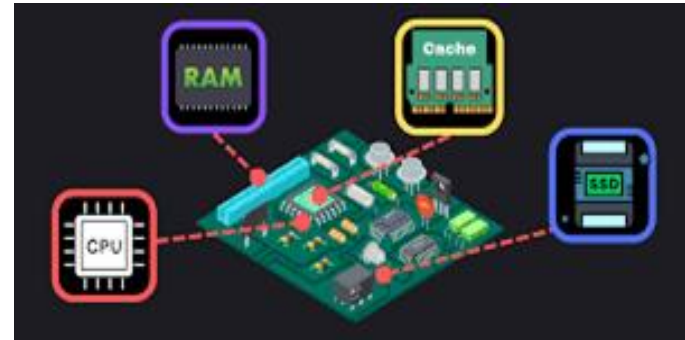
Computer Architecture:

Computer architecture defines how a computer's components communicate through electronic signals to perform input, processing, and output operations.

- It covers the design and organization of the CPU, memory, storage, and input/output devices.
- Describes how these components interact through buses, control signals, and data pathways.
- It directly influences the overall speed, functionality, and reliability of a computer system.



Basic Architecture of a Computer



Different Components

Input Unit: How Does an Input Device Work?

An input device converts user actions or physical movements into signals that the computer can understand and process.

The input unit consists of input devices that are attached to the computer.

These devices take input and convert it into binary language that the computer understands.

- **Signal Conversion:** Input devices convert physical actions (e.g., pressing a key, moving a mouse) into electrical signals that the computer can understand.
- **Data Transmission:** These electrical signals are transmitted to the computer's processor, typically via USB, Bluetooth, or wireless connections.

- **Processing by the CPU:** Once the signals reach the processor, the data is interpreted and processed by the computer to carry out the desired task (e.g., typing a letter, moving a cursor).
- **User Interaction:** The user interacts with the input device (keyboard, mouse, etc.), and the device sends corresponding data to the computer based on user input.
- **Feedback to the User:** In some cases, input devices provide feedback (e.g., vibration in a game controller or sound from a keyboard key press) to confirm the input was registered.

Input Devices

www.storyplanets.com



Keyboard



Mouse



Touchscreen



Scanner



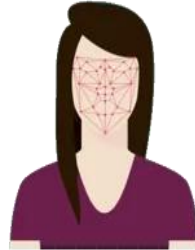
Microphone



Webcam



Joystick



Biometric Device



Barcode Reader



Stylus

Forensic Importance of Input Devices/Units:

Input devices capture user actions → which can provide **behavioral, biometric, or operational evidence**.

1. Keyboard

- **Keyloggers** (hardware/software) may be installed, giving insight into stolen credentials.
- Residual data (buffer memory) could contain recent keystrokes.
- Wear analysis (forensic inspection of worn-out keys) can hint at commonly typed passwords.

2. Mouse / Touchpad

- Movement patterns can indicate user activity (timestamps, session usage).
- Biometric behavior (mouse dynamics, click timing) can help identify a specific user.

3. Touchscreens (mobiles, tablets, kiosks)

- Touch coordinates, swipe patterns, and pressure data may be logged.
- Fingerprints and smudge patterns (physical forensics) can indicate PINs/patterns.

1. Microphones

- Audio recordings may be intercepted or stored (e.g., cached by apps).
- Can provide **voice evidence**, background noise clues, or indicate unauthorized surveillance.

2. Cameras / Scanners

- May hold cached photos or scanned documents.
- Metadata (EXIF: time, GPS, device ID) can provide location + usage info.

3. Biometric Devices (fingerprint scanners, iris scanners)

- Stored biometric templates can confirm user identity.
- Logs show authentication attempts (success/failure timestamps).

Output Unit

An output unit of a computer system performs the following functions:

1. It accepts the results produced by the computer, which are in coded form and hence, cannot be easily understood by us
2. It converts these coded results to human acceptable (readable) form
3. It supplies the converted results to outside world

Output Devices

Output Devices				www.storyplanets.com
				
Monitor	Printer	Speakers	Headphone	
				
Projector	Plotter	Braille Reader	Haptic Feedback Device	

Forensic Importance of Output Devices/Units:

Output devices reveal **what was shown, printed, or transmitted to a user**, which can be crucial for **reconstructing actions**.

1. Monitors / Displays

- **Screen recording logs / screenshots** may be cached (esp. in VMs, RDP, or apps).
- Pixel residue (frame buffer memory) can retain parts of displayed content after shutdown.

2. Printers

- **Print spool files** (on host computer) reveal documents printed.
- Internal printer memory can cache recent jobs.

1. Speakers / Headphones

- May not store data, but forensic analysis of connected system logs can prove playback (e.g., specific audio played).
- Can be abused as microphones (side-channel attacks).

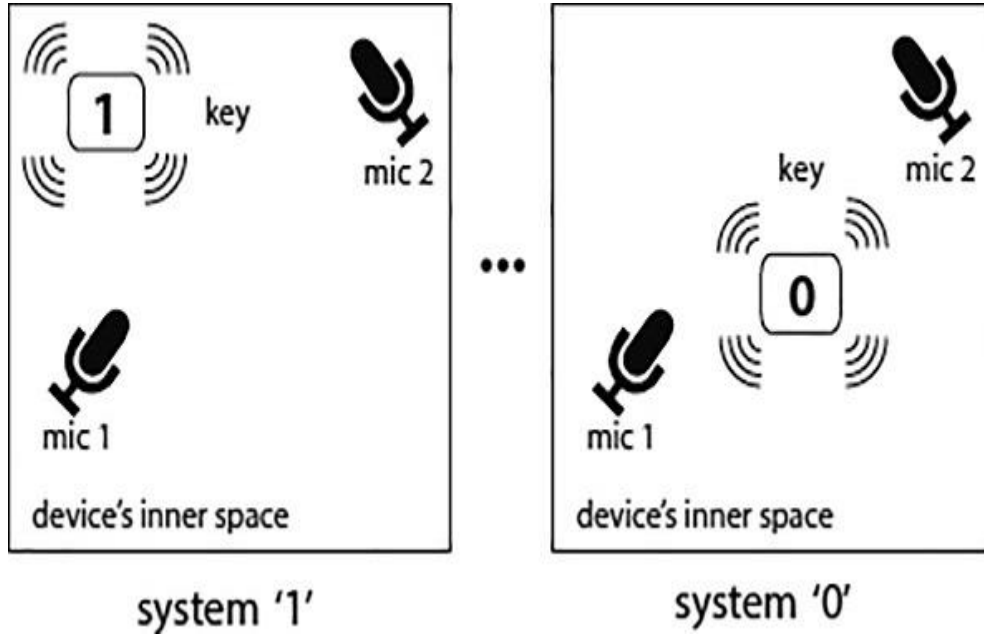
2. Projectors / External Displays

- Logs may store device connection history.
- Cached presentation files in associated software.

3. Haptic Devices (VR, controllers, feedback systems)

- Interaction logs can confirm user presence and activity in virtual environments.

Eg:



Acoustic Side-Channel attack
on keyboard

Central Processing Unit (CPU) :

Often referred to as the "brain" of the computer, the CPU executes instructions, performs calculations, and manages data.

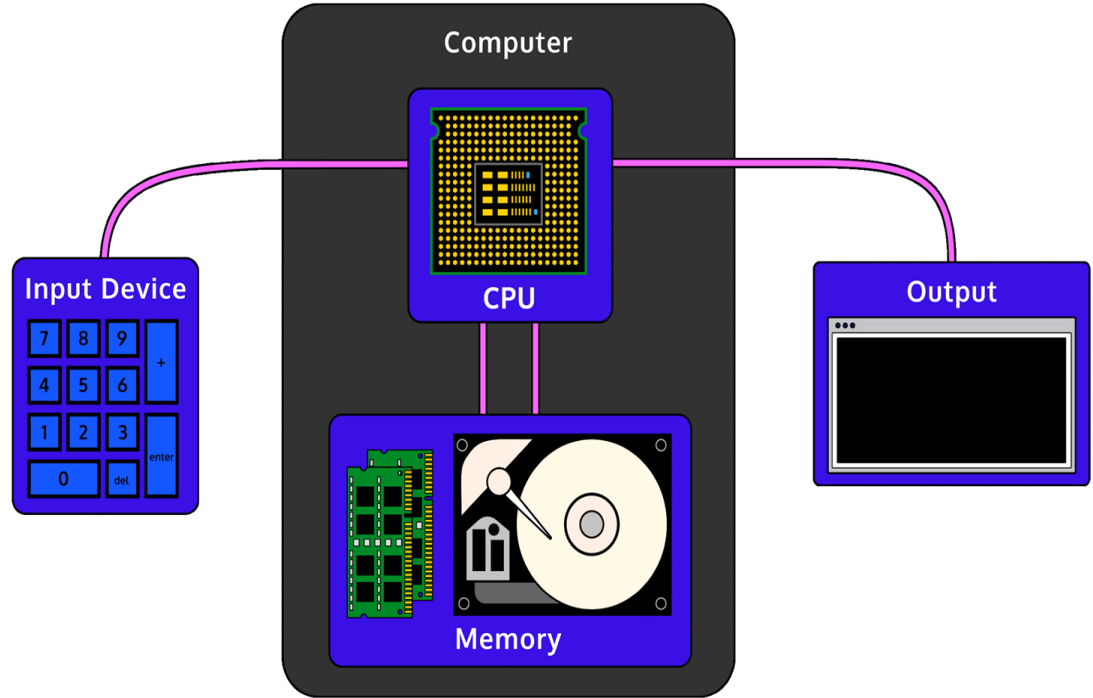
Its architecture dictates factors such as instruction set, clock speed, and cache hierarchy, all of which significantly impact overall system performance.

The CPU is usually placed in a special slot called a **socket** on the computer's **motherboard**, which is like the main circuit board that connects all the parts of a computer. The CPU handles tasks like:

- Doing math **calculations** (like adding or multiplying numbers) and **comparisons**.
- Running apps or games.
- Helping the keyboard, mouse, and screen work together.
- Storing and retrieving information during tasks.

Main Components

- **Control Unit (CU):** “Traffic police” → directs data flow between CPU, memory, and devices.
- **Arithmetic Logic Unit (ALU):** “Calculator” → performs math (+, -, *, /) and logic (AND, OR, NOT, comparisons).
- **Registers:** “Scratchpad” → very small, very fast memory inside the CPU. Hold instructions, operands, results.
- **Cache:** “Short-term memory” → faster than RAM, stores recently/frequently used data.
- **Clock:** synchronizes operations in cycles (GHz = billions of ticks per second).



More content (just read)

- **Control Unit:** The control unit manages the CPU by sending signals like clock, hold, and reset to its parts. It ensures all components work together to complete tasks. For example, it synchronizes data movement from cache memory to the ALU.
- **Arithmetic and Logic Unit (ALU):** The ALU handles arithmetic tasks (like addition, subtraction, multiplication, division) and logical tasks (like AND, OR, comparisons). It uses addition for all calculations, e.g., solving 2×3 as $2 + 2 + 2 = 6$.
- **Memory Unit:** The memory unit stores data and instructions. Older CPUs used registers, but modern ones also have fast cache memory. The CPU fetches data from RAM, ROM, or hard disks and stores it in registers or cache during tasks.

Major components of CPU

Control Unit:

- Acts as the central nervous system of a computer system
- Selects and interprets program instructions, and coordinates execution
- Has some special purpose registers and a decoder to perform these activities

Arithmetic and Logic Unit:

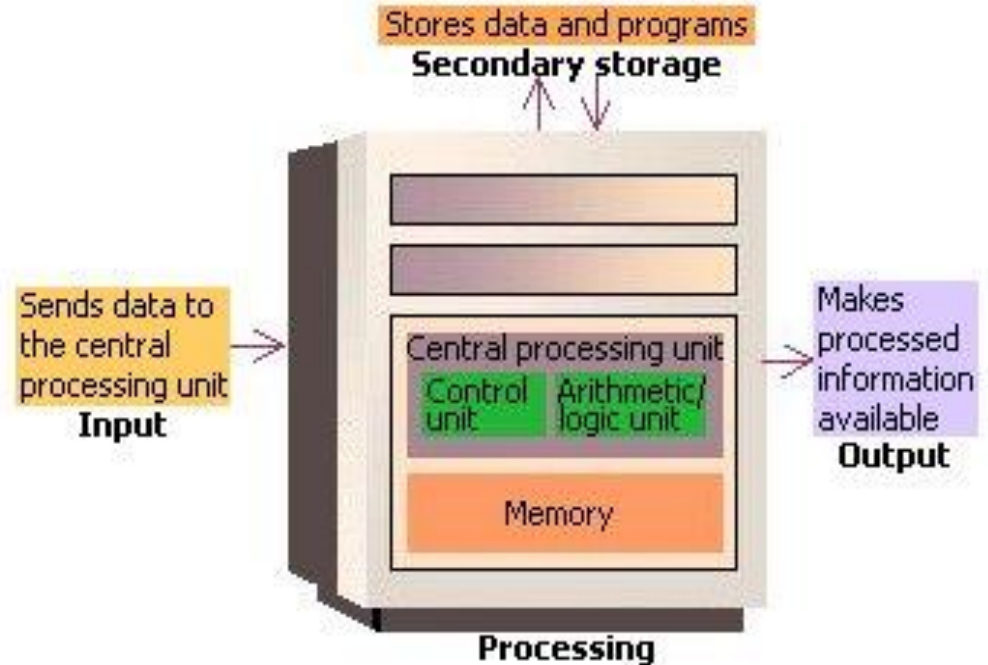
- Actual execution of instructions takes place in ALU
- Has some special purpose registers
- Has necessary circuitry to carry out all the arithmetic and logic operations included in the CPU instruction set

Core Idea

The CPU = the **brain of the computer**. Its job is to:

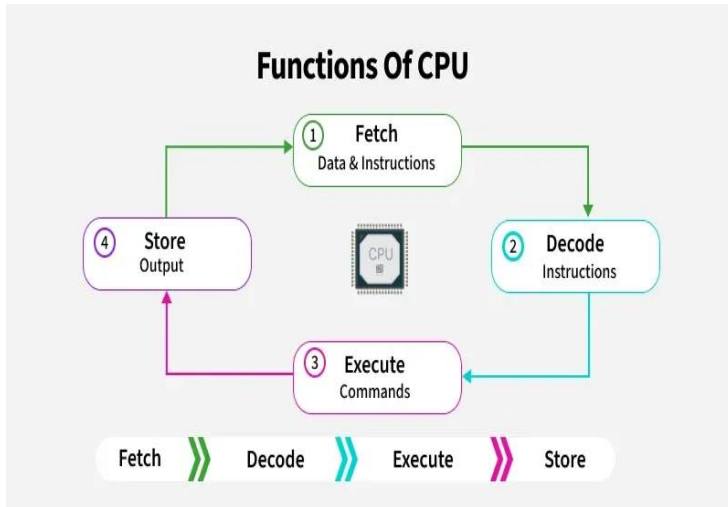
- **Fetch** instructions from memory
- **Decode** them (understand what to do)
- **Execute** them (perform calculation or action)
- **Store** the result back

This cycle repeats billions of times per second = **Instruction Cycle (Fetch–Decode–Execute Cycle)**.



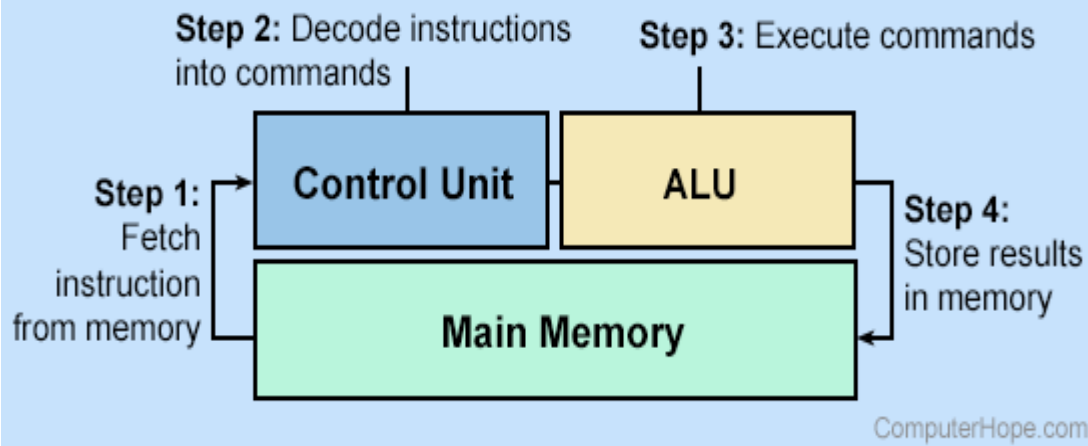
Functions of the CPU

The CPU's main job is to process instructions from programs. It does this through a process called the **Fetch-Decode-Execute-Store** cycle:



1. **Fetch:** the first CPU gets the instruction. That means binary numbers that are passed from RAM to CPU.
2. **Decode:** When the instruction is entered into the CPU, it needs to decode the instructions. with the help of ALU(Arithmetic Logic Unit), the process of decoding begins.
3. **Execute:** After the decode step the instructions are ready to execute.
4. **Store:** After the execute step the instructions are ready to store in the memory.

Machine Cycle



Instruction Set Architecture

- CPU has **built-in ability** to execute a particular set of machine instructions, called its instruction set
- Most CPUs have 200 or more instructions (such as add, subtract, compare, etc.) in their instruction set
- CPUs made by different manufacturers have different instruction sets
- Manufacturers tend to group their CPUs into “families” having similar instruction sets
- New CPU whose instruction set includes instruction set of its predecessor CPU is said to be backward compatible with its predecessor.

Registers

Special memory units, called registers, are **used to hold information** on a **temporary basis** as the **instructions are interpreted and executed by the CPU**

Registers are part of the CPU (not main memory) of a computer

The length of a register, sometimes called its word size, equals the number of bits it can store

Some commonly used Registers

Sr. No.	Name of Register	Function
1	Memory Address (MAR)	Holds address of the active memory location
2	Memory Buffer (MBR)	Holds contents of the accessed (read/written) memory word
3	Program Control (PC)	Holds address of the next instruction to be executed
4	Accumulator (A)	Holds data to be operated upon, intermediate results, and the results
5	Instruction (I)	Holds an instruction while it is being executed
6	Input/Output (I/O)	Used to communicate with the I/O devices

Forensic Importance:

- Can reveal **current instructions being executed** at the time of seizure.
- May hold **encryption keys, passwords, or memory addresses** actively used.
- Evidence of **running malware/rootkits** (e.g., control flags, jump addresses).
- **Very volatile** → useful mainly in live forensics (must capture before shutdown).

Where to Find ?

Inside the CPU chip itself (not directly accessible by normal tools).

- Capturable only via **live memory acquisition** tools (e.g., Volatility plugins).

When to Capture:

- **Immediately during live forensics** (before shutdown).
- Evidence Lost once the system powers off or reboots.

Evidence Examples: Running instruction, program counter (next instruction), crypto keys loaded in registers.

Processor Speed

Computer has a built-in system clock that emits millions of regularly spaced electric pulses per second (known as clock cycles)

It takes one cycle to perform a basic operation, such as moving a byte of data from one memory location to another

Normally, several clock cycles are required to fetch, decode, and execute a single program instruction

Hence, **shorter the clock cycle, faster the processor**

Clock speed (number of clock cycles per second) is measured in Megahertz (10^6 cycles/sec) or Gigahertz (10^9 cycles/sec)

Types of CPUs

CPUs come in different types, depending on how many **cores** they have. A core is like a mini-CPU inside the main CPU, and more cores mean the CPU can do more tasks at once. Here are the main types:

- **Single-Core CPU:** The oldest type, used in the 1970s. It can only handle one task at a time, so it's slow for modern apps like games or web browsers.
- **Dual-Core CPU:** Has two cores, so it can handle two tasks at once. It's faster and better for multitasking, like listening to music while doing homework.
- **Quad-Core CPU:** Has four cores, making it great for heavy tasks like video editing or playing modern games. It's very fast and common in today's computers.

Storage Unit

The storage unit of a computer system holds (or stores) the following :

1. Data and instructions required for processing (received from input devices)
2. Intermediate results of processing
3. Final results of processing, before they are released to an output device

Types of Storage:

Main / Primary Storage Features:

- Every computer has a temporary storage built into the computer hardware
- It stores instructions and data of a program mainly when the program is being executed by the CPU.
- Used to hold running program instructions
- Used to hold data, intermediate results, and results of ongoing processing of job(s)
- This temporary storage is known as main memory, primary storage, or simply memory.
- Volatile (loses data on power dissipation)
- Fast in operation ; Small Capacity
- Expensive

RAM

Primary storage of a computer is often referred to as RAM because of its random access capability

RAM chips are volatile memory

A computer's motherboard is designed in a manner that the memory capacity can be enhanced by adding more memory chips

The additional RAM chips, which plug into special sockets on the motherboard, are known as single-in-line memory modules (SIMMs)

Forensic Uses of RAM

1. Running Processes & Programs

- RAM holds **active applications and OS processes**.
- Investigators can see **what programs were running at capture time**.
- Example: Malware running only in memory (“fileless malware”) leaves traces here.

Thus, Identify suspicious processes, injected code, hidden programs.

2. Open Network Connections

- RAM stores details of **network sockets, IP addresses, ports**.
- Example: A backdoor malware connecting to attacker’s server.

Detect exfiltration, active command-and-control channels.

3. Passwords & Encryption Keys

- Many programs temporarily load **usernames, passwords, API keys, cryptographic keys** into memory.
- Example: Browser session cookies, VPN credentials, or disk encryption keys.

Recover authentication data to access encrypted evidence.

4. Unsaved Documents & Data

- Anything unsaved (Word files, chat messages, clipboard contents) exists in RAM.
- Example: A suspect typed a ransom note in Notepad but never saved it — still retrievable from RAM dump.

Capture data that never touches disk.

5. System Artifacts

- RAM contains OS-level structures:
 - Registry hives (cached)
 - Loaded drivers
 - Kernel modules
 - Memory-mapped files

Detect rootkits, injected DLLs, or persistence mechanisms.

6. Anti-Forensic / Malware Evidence

- Advanced malware resides only in RAM to avoid disk traces.
- Example: Banking Trojans, ransomware loaders, in-memory exploits.

Only a **live memory capture** reveals them.

ROM: Read Only Memory

ROM a non-volatile memory chip

Data stored in a ROM can only be read and used – they cannot be changed

ROMs are mainly used to store programs and data, which do not change and are frequently used.

For example, system boot program

Forensic Importance:

- **Firmware analysis:** Identify tampering or rootkits hidden in BIOS/UEFI.
- Extract **device identifiers** (serials, IMEI, MAC addresses).
- Recover **boot sequence information** → was the system altered to hide activity?
- Useful for validating **trusted boot** (e.g., Secure Boot forensic checks).

Where to Find:

- Non-volatile chip on the motherboard (BIOS/UEFI flash memory).
- Can be read using:
 - Firmware dump tools (e.g., CHIPSEC, Flashrom).

When to Capture:

- Anytime — survives shutdown and reboot.
- Best done during **post-mortem analysis**.

Evidence Examples: Rootkits in UEFI, altered Secure Boot keys, hidden persistence mechanisms.

(Additional info)

Type	Usage
Manufacturer-programmed ROM	Data is burnt by the manufacturer of the electronic equipment in which it is used.
User-programmed ROM or Programmable ROM (PROM)	The user can load and store "read-only" programs and data in it
Erasable PROM (EPROM)	The user can erase information stored in it and the chip can be reprogrammed to store new information

Type	Usage
Ultra Violet EPROM (UVEPROM)	A type of EPROM chip in which the stored information is erased by exposing the chip for some time to ultra-violet light
Electrically EPROM (EEPROM) or Flash memory	A type of EPROM chip in which the stored information is erased by using high voltage electric pulses

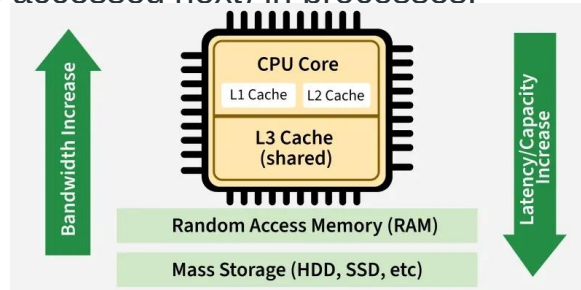
Cache memory

Cache memory is a small, high-speed storage area in a computer.

It stores copies of the data from **frequently used main memory locations**.

There are various independent caches in a CPU, which store instructions and data.

- The most important use of cache memory is that it is **used to reduce the average time to access data from the main memory**.
- The concept of cache works because there exists locality of reference (the same items or nearby items are more likely to be accessed next) in processes.



Cache Memory

It is commonly used for minimizing the memory processor speed mismatch.

It is an extremely fast, small memory between CPU and main memory whose access time is closer to the processing speed of the CPU.

It is used to **temporarily store very active data and instructions during processing.**

Forensic Importance of CPU

1. Control Unit (CU)

- Directs the flow of instructions and data between CPU, memory, and I/O.
- **Forensic importance:**
 - Determines **execution flow** → can reveal how malware or malicious code hijacks instruction flow (e.g., jump to shellcode).
 - Abnormal execution traces (found via memory forensics or debugging) can show process injection or tampering.
 - Analysis of instruction flow helps reconstruct **what program was running at a given time**.

2. Arithmetic Logic Unit (ALU)

- Performs arithmetic (+, −, ×, ÷) and logical operations (AND, OR, comparisons).
- **Forensic importance:**
 - Useful in **reverse engineering malware** – analyzing how malicious code manipulates values (e.g., encryption routines, obfuscation).
 - Crypto-malware (ransomware) relies heavily on ALU operations; understanding ALU-level execution helps detect **encryption keys or routines**.

3. Registers

- Small, high-speed storage directly inside CPU. Examples: Instruction Register (IR), Program Counter (PC), Accumulator, General-purpose registers.
- **Forensic importance:**
 - Contain **the most recent instructions and operands** during execution.
 - A memory dump may reveal sensitive data like passwords, cryptographic keys, or shellcode fragments in registers.
 - **Program Counter (PC)** shows *which instruction was executing at crash or seizure time* — key for reconstructing attacks.
 - Timing analysis of cache usage can reveal patterns of malicious activity.

4. Cache (L1, L2, L3)

- Very fast memory close to CPU, stores recently/frequently used instructions/data.
- **Forensic importance:**
 - **Cache residue** may contain fragments of sensitive data (passwords, cryptographic material, portions of files).
 - Side-channel attacks (like **Spectre**, **Meltdown**) exploit cache behavior to leak secret data.

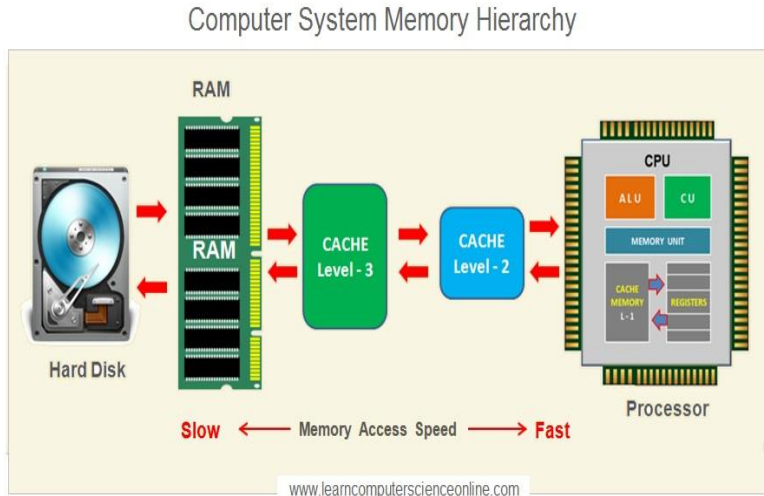
5. Clock

- Provides timing signals that synchronize CPU operations.
- **Forensic importance:**
 - CPU clock speed can affect **timestamp reliability** in logs.
 - Clock drift analysis helps in **event timeline reconstruction** during forensic investigation.

6. Instruction Set Architecture (ISA)

- Defines the CPU's "language" (x86, ARM, RISC-V).
- **Forensic importance:**
 - Determines **how malware is written and executed** (x86 shellcode vs ARM payloads).
 - Essential when performing **binary analysis** — forensic tools must match ISA to correctly interpret instructions.
 - Evidence from IoT/OT devices often requires ARM/RISC-V disassembly, not x86.

Memory Hierarchy:



- This includes various types of memory, such as cache memory, random access memory (RAM), and storage devices.
- The memory hierarchy plays a crucial role in optimizing data access times, as data moves between different levels of memory based on their proximity to the CPU and the frequency of access.

Conclusion

Input devices = what the user did.

Output devices = what the system showed.

Both can leave **forensic traces** in memory, logs, or caches.

Must be analyzed alongside **storage media** for complete evidence.